Adaptive Dashboards & Backend Foundations for Progress Analytics

Introduction

This merged research document integrates two key areas: 1. Adaptive learner-facing dashboards and user experience principles. 2. Backend infrastructures that support progress analytics, data tracking, and personalized feedback.

Together, these provide a foundation for designing adaptive educational applications that personalize learning, visualize growth, and utilize data driven decision making.

Adaptive Learning Analytics Dashboards

Research on Learning Analytics Dashboards (LADs) highlights how adaptive dashboards help consolidate learner progress and behavioral data.

Key Insights

- Dashboards increase awareness and reflection when aligned with pedagogical goals.
- Non-adaptive dashboards may reduce learning effectiveness.
- Interfaces should be customizable at an individual level.
- LADs should promote student–student and student–instructor interaction.
- Dashboards support time management, planning, and metacognitive processes.
- Adaptive features (personalized recommendations, alerts, transparency) improve self-regulation.
- Zimmerman's Self-Regulated Learning (SRL) model is often referenced in LAD design as it supports reflection, goal-setting, and effort monitoring.

Example Use Cases

- Moodle-style modular dashboards that show progress, completion, and resource history.
- Khan Academy–style adaptive dashboards (e.g., student progress rings, skill breakdowns).
- Game-inspired dashboards (e.g., Dokkan Battle profile screens) showing levels, stats, visual growth indicators, and progressive unlocks.

Adaptive UI/UX Design Principles

Core Principles

- **Device-specific layouts**: tailor visuals for mobile, tablet, and desktop.
- **Context-aware UI**: adapt the interface based on user behavior or task.
- **Performance-driven design**: load only necessary assets to increase efficiency.

Benefits

- Faster loads and reduced clutter.
- Improved accessibility across devices.
- Increased user retention from tailored experiences.
- Applicable to e-commerce, SaaS, media platforms, and personalized educational applications such as pronunciation coaching.

Adaptive UX Principles

Progressive Disclosure

A UX technique where advanced features remain hidden until the user needs them.

Goals

- Reduce cognitive overload.
- Ensure new users have a smooth learning path.
- Support both beginners and experts with scalable complexity.

Benefits

- Clearer, less crowded interfaces.
- Faster task completion.
- Reduced abandonment caused by complexity.

Tiered Information / Unlocking Features

Information appears in layers of complexity:

Examples

- Websites like Hacker News unlock capabilities as user reputation increases.
- In educational apps, completing lessons may unlock avatars, analytics, badges, or advanced challenge modes.

Benefits

- Rewards user engagement.
- Introduces complexity at appropriate stages.
- Encourages long-term platform use.

Backend Foundations for Progress Analytics

The second research component evaluates backend systems to support lesson tracking, assessment data, and personalized analytics.

Primary technologies explored include: - **xAPI + Learning Record Stores** - **Learnosity Analytics API** - **Adobe Learning Manager API** - **Supporting databases** such as Supabase, Firestore, Appwrite, and Hasura.

1. xAPI + Learning Record Store (LRS)

A standards-based method for capturing learning experiences using structured statements.

Benefits

- Flexible statements: supports events such as “started,” “skipped,” “progressed,” or “completed.”
- Portability: LRSs can share learning histories between systems.
- Works across devices and allows offline/low-connectivity activity logging.
- Strong option for personalized learning due to granular event data.

Relevant LRS Platforms

- Learning Locker (open source)
- Yet Analytics LRS
- SCORM Cloud LRS
- Veracity Learning LRS

Flutter/Dart Aspect

No official SDK; developers send HTTPS JSON statements via http/dio.

2. Learnosity (Analytics API)

Benefits

- Detailed assessment reporting (correctness, attempts, time).
- Completion tracking across multi-activity journeys.

- Embeddable analytics: item-level, session-level, and standards-based reports.
- Supports recommendation engines through skill-weakness identification.

Cons

- Commercial platform (not open source).

3. Adobe Learning Manager (ALM) API

Benefits

- OAuth2 security and scoped API tokens.
- Progress endpoints (loResourceGrades) providing duration, progress percentage, pass/fail, and scores.
- Structured API tools (fields, include, pagination) for efficient queries.
- Useful for learner progress, completions, enrollments, and analytics reporting.

Supporting Databases

- **Supabase (Postgres)** – relational, row-level security, edge functions.
- **Firebase Firestore** – document DB for real-time updates.
- **Appwrite** – backend-as-a-service with RLS.
- **Hasura** – instant GraphQL API over Postgres.

Data Tracking Methods

Client-Side Tracking

Browser sends data directly to servers or analytics endpoints via tags/pixels. Common in marketing and product analytics.

Server-Side Tracking

Data is sent first to a server, which then forwards it to other systems. Improves security, reduces ad blockers, provides greater reliability.

Example Implementations

Adobe Learning API Analytics → Frontend

A dashboard visualizes: - completion rates - enrollment numbers - learner statuses - compliance progress

Learnosity API Analytics → Frontend

Embedded reports show detailed skill performance, session history, and learner-level summaries.

Preliminary Data Model Sketches

Example 1: General Learning Progression Schema

Includes tables for: - users - courses and modules - lessons and assessments - progress_events - assessment_results - recommendations

(See page diagrams in the PDF.)

Example 2: Pronunciation Coach Preliminary Schema

A more detailed schema including: - phoneme-level scoring - session metadata - pronunciation attempts - feedback events - personalized recommendation pipelines

References

Full reference lists provided in the original documents.